



# CSE 250

## Circuits and Electronics

Undergraduate Program

### CSE250: Circuits and Electronics

Offered:

Soft Pre-requisite: PHY112

Fundamental electrical concepts and measuring units of electrical charge, voltage, current, resistance, and power; Laws of electricity (Ohm's law, Kirchhoff's Current and Voltage law) and various methods of electrical circuit analysis (Nodal, Mesh); Introduction to basic electrical circuit elements; I-V characteristics; Circuit analysis in Direct current, First-order Transient and Alternating current mode, for various combinations of Resistive, Inductive and Capacitive networks; Phasor representation of sinusoidal quantities; Circuit theorems for linear circuits (Source Transformation, Superposition, Thevenin, Norton and Maximum Power Transfer). This course includes compulsory 3-hour laboratory work.

#### Course Objectives

The objectives of this course are to:

1. Introduce students to ideal linear electrical circuit components such as dependent and independent voltage and current sources, resistors, capacitors, and inductors and their characteristic equations.
2. Illustrate the I-V characteristics of any two-terminal devices and infer circuit equivalence.
3. Define physical quantities related to electricity such as voltage, current, and power, and introduce passive sign convention for computing these quantities.
4. Explain fundamental laws like Ohm's law, and Kirchhoff's voltage and current law, as well as important linear circuit theorems such as Thevenin's and Norton's theorem, maximum power transfer theorem, superposition principle, and source transformation.
5. Familiarize students with several circuit-solving techniques aside from the circuit theorems, such as the voltage/current divider rule, series-parallel circuit equivalence, and nodal and mesh analysis, that take advantage of the fundamental laws and theorems of the linear circuit.
6. Analyze first-order transient circuits with resistors, capacitors, and inductors in the time domain.
7. Introduce phasors and analyze alternating current (AC) circuits constructed from sinusoidal sources, resistors, capacitors, and inductors in the phasor domain.

#### List of Books

1. Fundamentals of Electric Circuits, Charles K. Alexander, Matthew N. O. Sadiku, 2019, 6th, McGraw Hill Education, 978-9353165505
2. Foundations of Analog and Digital Electronic Circuits, Anant Agarwal, Jeffrey H. Lang, 2005, 1st, Morgan Kaufmann Publishers, 978-1558607354
3. Electric Circuits, James W. Nilsson, Susan A. Riedel, 2010, 9th, Pearson College Div, 978-0136114994
4. Introductory Circuit Analysis, Robert L. Boylestad, 2013, 12th, Pearson Education India, 978-9332518612

#### Course Materials

1. Lecture slides
2. Practice Problems
3. Video lectures
4. Lab handouts
4. Simulation tools

#### Course Outcome

#### Course Coordinator



Mr. Shadman Shahriar



Mr. Purbayan Das

#### Lab Coordinator



Md. Saiful Bari Siddiqui

| #   | DESCRIPTION                                                                                                                                                                                                                                                | WEIGHT |                              |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------|
| CO1 | Understand and Describe the foundational concepts of electricity, including relevant physical quantities and the governing laws that dictate its behavior, such as Kirchhoff's current and voltage law, and Ohm's law, etc.                                | 10     | <a href="#">VIEW DETAILS</a> |
| CO2 | Describe linear circuit theorems, such as the superposition principle, source transformation, Thevenin and Norton's theorem, and maximum power transfer theorem, and demonstrate the ability to Apply them efficiently.                                    | 35     | <a href="#">VIEW DETAILS</a> |
| CO3 | Analyze the behavior of analog electrical circuits constructed from networks of diverse linear elements by utilizing various tools, including nodal and mesh analysis, circuit equivalence, voltage and current divider rules, and phasors domain analysis | 35     | <a href="#">VIEW DETAILS</a> |
| CO4 | Demonstrate competence in using laboratory equipment, such as oscilloscopes, function generators, and multimeters, to build, test, and verify analog circuits, and troubleshoot circuit problems.                                                          | 10     | <a href="#">VIEW DETAILS</a> |
| CO5 | Collaborate effectively in a group in the laboratory, and Report their findings and insights clearly and concisely, using technical language and documentation standards.                                                                                  | 4      | <a href="#">VIEW DETAILS</a> |
| CO6 | Design schematics and Simulate electrical circuits using software programs, such as LTspice and EveryCircuit to analyze circuit behaviors.                                                                                                                 | 4      | <a href="#">VIEW DETAILS</a> |

Course Lectures

| WEEK    | LECTURE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO MAP                                     |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| Week 1  | Illustrating the motivation behind taking this course. What are the real-life implications of these course materials? Discuss basic circuit parameters like voltage, current, energy, and power definitions and units. Introducing passive sign convention, positive-negative voltage/current/power. Discuss different types of circuit elements (active, passive), and different types of sources (DC/AC, voltage/current, dependent/independent). Introducing circuit symbols.                                                                                                                                                                                                                                                                                                                                                                                                                           | <a href="#">CO1</a>                        |
| Week 2  | Introducing basic electrical components: resistors, voltage source, and current source. Basic laws of electrical circuits: Ohm's law. I-V characteristics of a resistor. Discuss passive sign convention, finding the power of circuit elements by P=VI. Defining nodes, loops, and mesh. Discuss various circuit configurations: Series, Parallel, and others. How to identify series and parallel connections and calculate equivalent resistance. Open and short circuit. Defining Node/Supernode. Introducing Current Sign Convention. Basic laws of electrical circuits: Kirchhoff's current law. Statement and application of KCL. Current divider rule in a parallel circuit. Illustrating convention doesn't change the KCL equation. Show the usefulness of Supernode.                                                                                                                            | <a href="#">CO1</a>                        |
| Week 3  | Defining Mesh/Supermesh. Revisiting Passive Sign Convention. Basic laws of electrical circuits: Kirchhoff's voltage law. Statement and application of KVL. Voltage divider rule in a series circuit. Illustrating the assumption of the current direction doesn't change the KVL equation. Show the usefulness of Supermesh. I-V characteristics of basic circuit elements: Resistor, Voltage source, Current source, Open circuit, Short circuit, any two-terminal device/circuit, a combination of elements (e.g. voltage/current source in series/parallel with resistor). The idea of circuit equivalence. Series-parallel equivalent circuit for resistance/voltage source/current source combinations. Ideal/non-ideal current/voltage source. Simplifying circuits by means of equivalence. Basic circuit theorem: Source Transformation theorem. Failure of applying in Wheatstone bridge circuit. | <a href="#">CO1</a><br><a href="#">CO2</a> |
| Week 4  | Explaining the Nodal Analysis technique, using it to solve for current, voltage, and power in a given circuit (multiple examples). Reintroducing dependent sources. Demonstrating Nodal Analysis with Dependent Sources. Problems with floating voltage sources, using Supernodes to solve such circuits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">CO3</a>                        |
| Week 5  | Explaining the Mesh Analysis technique, using it to solve for current, voltage, and power in a given circuit (multiple examples). Demonstrating Mesh Analysis with dependent sources. Problems with common current sources, using Supermeshes to solve such circuits.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <a href="#">CO3</a>                        |
| Week 6  | Linear circuit elements. Linearity of voltage, current in circuits, and non-linearity of power. Circuit theorem: Superposition theorem. Using superposition theorem for solving DC circuits. Superposition Theorem for circuits with Dependent Sources. Reintroduction to circuit linearity, I-V characteristics of linear circuits. Circuit Theorems: Thevenin's theorem. The motivation behind Thevenin's theorem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">CO2</a>                        |
| Week 7  | Using test voltage/current sources while deactivating sources to find Thevenin's. Solving resistance matching problems for transferring maximum power. Norton's theorem, the relation between Thevenin's and Norton's theorem. Using Thevenin's/Norton's theorem for solving circuits. Maximum transferable power and condition for it.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">CO2</a>                        |
| Week 8  | Capacitors and Inductors, their component equations. SI unit for measuring capacitance and inductance. Transient circuits, visualizing and analyzing transient circuits. Response of transient circuit: first-order RC circuit, time constant. Analyzing and plotting first-order transient circuit response. Finding capacitor current from capacitor voltage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <a href="#">CO1</a><br><a href="#">CO3</a> |
| Week 9  | Response of transient circuit: first order RL circuit, time constant. Analyzing and plotting first-order transient circuit response. Finding inductor voltage from inductor current. Complex number review. Alternating current, the importance of AC circuit. Visualizing the dynamics of an AC circuit, Amplitude, and RMS voltage/current and finding them from a graph.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">CO1</a><br><a href="#">CO3</a> |
| Week 10 | Introducing Impedance. Defining impedance for various elements, Phasor analysis of an AC circuit. Instantaneous voltage, current, and power. Applying the superposition theorem on AC circuits containing sources of different frequencies                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">CO3</a>                        |

Lab Lectures

| WEEK   | LECTURE                                                                                                                                                                                 | CO MAP              |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Week 1 | Introduction to Laboratory Instruments (Part 1). Introduction to Series and Parallel Circuits.                                                                                          | <a href="#">CO4</a> |
| Week 2 | Verification of KVL and KCL. Verification of Superposition Principle.                                                                                                                   | <a href="#">CO4</a> |
| Week 3 | Introduction to Laboratory Instruments (Part 2). I-V Characteristics and Circuit Equivalence.                                                                                           | <a href="#">CO4</a> |
| Week 4 | Study of I-V Characteristics, and Verification of Thevenin's and Maximum Power Theorem using Software (LTspice) Simulation. (Dependent sources, I-V characteristics, sweep parameters). | <a href="#">CO6</a> |
| Week 5 | Verification of Thevenin's Theorem and Maximum Power Transfer Theorem.                                                                                                                  | <a href="#">CO4</a> |
| Week 6 | Study of Transient Behaviour of RC Circuit.                                                                                                                                             | <a href="#">CO4</a> |
| Week 7 | Study of Transient Circuits Using Software Simulation. Study of AC Circuits Using Software Simulation.                                                                                  | <a href="#">CO6</a> |

